

COURSE: MATH 251 — Probability and Statistics for CS

Credits: 3 | Department: Mathematics

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PREREQUISITES: MATH 152

CO-REQUISITES: CS 201 (must be taken concurrently or already completed)

OFFERED: Fall, Spring

DESCRIPTION:

Probability and statistics with applications to computer science. Topics include sample spaces, conditional probability, Bayes theorem, discrete and continuous random variables (Bernoulli, binomial, Poisson, normal, exponential), expectation, variance, covariance, law of large numbers, central limit theorem, hypothesis testing, confidence intervals, and maximum likelihood estimation. Essential prerequisite for CS 351 (Machine Learning).

GRADING: Weekly problem sets 40%, Midterm 30%, Final 30%.

NOTE: CS 201 must be taken concurrently or already completed. The co-requisite exists because probability assignments use programming simulations.